SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction , for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding ***(2)***	Method Selection ***(2)***	Solution Process ***(2.5)***	Accuracy of Calculation ***(2)***	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I am B V Chaitanya Reddy-192425376 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Annexure B
Analytical Problem Solving

1. Covariance Analysis in Health Insurance (10 Marks)

The table below shows the **annual premium increase rate (xi)** and the **number of claims per 100 policyholders (yi)** in a health insurance company.
Using the covariance formula, determine whether premium increase and claim frequency have a **positive or inverse relationship**.
Before computing covariance, calculate the **mean of x and y**.
Premium Increase % (xi) Claims per 100 Policyholders (yi)

3.2 15 4.0 18 5.5 25

4.8 20

---

2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analyzing **patient blood pressure readings (systolic)** to identify trends. Due to minor fluctuations, the analyst performs **local smoothing using equal-frequency (equi-depth) binning**.
Sample Data Set (Systolic BP readings):
110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

---

3. Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates **Body Mass Index (BMI)** of 18 clients for risk classification. To simplify analysis, they smooth the data by replacing values with **closest bin boundaries using equal-width partitioning**. **Sample Data Set (Sorted BMI):**
18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

---

4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the **cost of medical consumables (in ₹)**. To reduce data for mining purposes, they discretize the sorted price records using **bin medians**.
Sample Data Set (Costs in ₹):

120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

---

5. Standardization of Patient Age Data (5 Marks)

Given the following **ages of insured individuals**:
18, 22, 25, 42, 28, 43, 33, 35, 56, 28
Standardize the variable by performing the following:
(a) Compute the **mean absolute deviation (MAD)** of age.
(b) Compute the **z-score for the first four observations**.

Solutions :

Q1. Covariance analysis in health insurance

Given data

Premium increase (x)	**Claims per 100 policyholders (y)**
3.2	15
4.0	18
5.5	25
4.8	20

We need to find the mean of x and y and then compute the covariance.

Step 1: Calculate the mean of x

Formula:

$$\bar{x} = \frac{\sum x_i}{n}$$

Here,

$$\sum x_i = 3.2 + 4.0 + 5.5 + 4.8 = 17.5$$

$$\bar{x} = \frac{17.5}{4} = 4.375$$

Therefore,

Mean premium increase = 4.375%

Step 2: Calculate the mean of y

Formula:

$$\bar{y} = \frac{\sum y_i}{n}$$

Here,

$$\sum y_i = 15 + 18 + 25 + 20 = 78$$

$$\bar{y} = \frac{78}{4} = 19.5$$

Therefore,

Mean number of claims = 19.5

Step 3: Prepare the covariance table

Covariance formula:

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n}$$

x	**y**	**x − 4.375**	**y − 19.5**	**(x−x̄)(y−ȳ)**
3.2	15	-1.175	-4.5	5.2875
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125

Step 4: Sum the products

$$5.2875 + 0.5625 + 6.1875 + 0.2125 = 12.25$$

Step 5: Calculate covariance

$$\text{Cov}(x, y) = \frac{12.25}{4} = 3.0625$$

Final answer

- Mean of x = 4.375
- Mean of y = 19.5
- Covariance = 3.0625

Interpretation

Since the covariance value is positive (3.0625), premium increase and claim frequency have a positive relationship.

This means that when the number of claims increases, the premium increase rate also tends to increase. In a health insurance company, higher claim frequency usually results in higher premiums to compensate for increased insurance costs.

Q2. Data smoothing using equal-frequency binning

Given data

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

There are 27 observations.

In equal-frequency binning, each bin contains the same number of observations.

Step 1: Create equal-frequency bins

Number of bins = 3

Observations per bin = 27 ÷ 3 = 9

Bin 1

110, 112, 115, 115, 118, 120, 120, 122, 124

Median (5th value) = 118

Bin 2

124, 126, 126, 126, 126, 130, 135, 135, 138

Median = 126

Bin 3

138, 138, 138, 140, 145, 150, 152, 160, 180

Median = 145

Step 2: Replace each value by the bin median

Smoothed Bin 1

118, 118, 118, 118, 118, 118, 118, 118, 118

Smoothed Bin 2

126, 126, 126, 126, 126, 126, 126, 126, 126

Smoothed Bin 3

145, 145, 145, 145, 145, 145, 145, 145, 145

Final smoothed data

118, 118, 118, 118, 118, 118, 118, 118, 118,

126, 126, 126, 126, 126, 126, 126, 126, 126,

145, 145, 145, 145, 145, 145, 145, 145, 145

Interpretation

Equal-frequency binning reduces small fluctuations in blood pressure readings and produces a smoother dataset. This helps in identifying overall patient trends without being affected by minor measurement variations.

Q3. Equal-width binning for BMI

Given data

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Step 1: Calculate the range

Minimum = 18.5

Maximum = 39.5

$$\text{Range} = 39.5 - 18.5 = 21$$

Step 2: Calculate bin width

Number of bins = 3

$$\text{Bin width} = \frac{21}{3} = 7$$

Step 3: Create equal-width bins

Bin	**Boundary**
Bin 1	18.5 – 25.5
Bin 2	25.5 – 32.5
Bin 3	32.5 – 39.5

Step 4: Replace values with the nearest boundary

Original BMI	**Nearest boundary**
18.5	18.5
19.2	18.5
21.8	18.5
24.9	25.5
25.5	25.5
26.2	25.5
26.4	25.5
27.8	25.5
29.2	32.5
30.2	32.5
30.4	32.5
31.9	32.5

32.4	32.5
33.1	32.5
33.6	32.5
34.7	32.5
38.2	39.5
39.5	39.5

Interpretation

Equal-width binning groups BMI values into uniform intervals. Replacing values with the nearest boundary simplifies the data and supports easier obesity risk categorization and insurance analysis.

Q4. Data reduction using bin medians

Step 1: Sort the data

120, 140, 145, 145, 148, 150, 150, 150, 155,

155, 155, 155, 160, 160, 165, 170, 170, 175,

175, 175, 175, 180, 190, 200, 210, 230, 300

Total observations = 27

Create 3 bins of 9 values each.

Bin 1

120, 140, 145, 145, 148, 150, 150, 150, 155

Median = 148

Bin 2

155, 155, 155, 160, 160, 165, 170, 170, 175

Median = 160

Bin 3

175, 175, 175, 180, 190, 200, 210, 230, 300

Median = 190

Replace each value with the median

Bin 1

148 repeated 9 times

Bin 2

160 repeated 9 times

Bin 3

190 repeated 9 times

Reduced dataset

148, 148, 148, 148, 148, 148, 148, 148, 148,

160, 160, 160, 160, 160, 160, 160, 160, 160,

190, 190, 190, 190, 190, 190, 190, 190, 190

Interpretation

Replacing values with bin medians reduces the number of distinct values while preserving the central tendency of each group. This is useful for data mining because it decreases storage requirements and improves processing efficiency.

Q5. Standardization of patient age data (10 marks)

Data:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

(a) Mean absolute deviation (MAD)

Step 1: Calculate the mean

$$\bar{x} = \frac{330}{10} = 33$$

Step 2: Calculate absolute deviations

Age	|Age − 33|
18	15
22	11
25	8
42	9
28	5
43	10
33	0
35	2
56	23
28	5

Sum = 88

$$\text{MAD} = \frac{88}{10} = 8.8$$

MAD = 8.8

(b) Z-score calculation

Step 1: Compute squared deviations

Age	Age − 33	(Age − 33)²
18	-15	225
22	-11	121
25	-8	64
42	9	81
28	-5	25

43	10	100
33	0	0
35	2	4
56	23	529
28	-5	25

Total = 1174

Step 2: Standard deviation

$$\sigma = \sqrt{\frac{1174}{10}} = \sqrt{117.4} = 10.84$$

Step 3: Z-scores

Formula:

$$z = \frac{x - \bar{x}}{\sigma}$$

Age	**Z-score**
18	-15 / 10.84 = -1.38
22	-11 / 10.84 = -1.02
25	-8 / 10.84 = -0.74
42	9 / 10.84 = 0.83

Final answer

- Mean = 33
- MAD = 8.8
- Standard deviation = 10.84
- Z-scores:
    - 18 → -1.38
    - 22 → -1.02
    - 25 → -0.74
    - 42 → 0.83

Interpretation

Standardization converts age values to a common scale. Negative z-scores indicate ages below the mean, while positive z-scores indicate ages above the mean. This helps compare patients of different age groups during insurance risk analysis.